

Photon Events in the Oyster Theory: State Update Dynamics Across the Hyperbrane

Author: Alberto Valis

1. Theoretical Context

In the framework of the Oyster Theory (OT), the photon is not treated as a discrete packet of energy or as a purely wave-like excitation propagating through spacetime. Instead, it is redefined as a state update operator acting within the Hyperbrane, the primary projection interface between the Bulk and the emergent Universe Bubble. In this model, the Bulk is an atemporal and aspatial informational medium where coherence states exist without metric constraints. The Hyperbrane functions as a resonance field, filtering and projecting coherent structures into the emergent spacetime. A photon event, in this context, represents a coherence update linking two distinct coordinates in the Hyperbrane projection.

2. Departure from Standard Physics

Classically, photons in Quantum Electrodynamics (QED) are quanta of the electromagnetic field, carrying energy and momentum and obeying the relativistic constraint of propagation at the speed of light c . In OT, however:

1) Bulk Perspective: The photon event is instantaneous — a simultaneous update of coherence state between two nodes (x_1, x_2) in the Hyperbrane projection.

2) Emergent Universe Perspective: The same event is observed as a finite-speed propagation, with an effective velocity $c_{\text{eff}} \leq c$, due to projection constraints and possible damping effects (e.g., interaction with dark matter distributions or Bulk resistance). This framework allows for the conceptual possibility of superluminal coordination in the Bulk while preserving relativistic causality in the emergent Universe Bubble.

3. Role in the Oyster Theory

Within OT, photon events are the primary messengers of coherence state changes between distant regions in the emergent Universe. They act as the operational link between the Super-Attractor (SA) — the gravitational-information pole in the Bulk — and the Hyperbrane, ensuring that local variations in resonance propagate throughout the coherent network. This redefinition is crucial because it positions the photon not as a carrier of energy through space, but as a trigger of state synchronization across the Hyperbrane's projection, allowing for the large-scale coherence necessary for Universe Bubble stability.

4. Implications

Velocity Reduction: If photon events are instantaneous in the Bulk but slowed in projection, then c_{eff} may be interpreted as the dissipative projection velocity, potentially influenced by dark matter distributions or Bulk inhomogeneities.

Observational Signatures: Small anomalies in photon arrival times across cosmological distances (beyond known dispersion measures) could hint at projection-related damping.

Testable Hypotheses: Correlations between high-energy photon delays and dark matter density maps could offer indirect support.

5. Mathematical Formalization

Let:

- x_1, x_2 : coordinates in the Hyperbrane projection
- t : emergent time coordinate in the Universe Bubble
- $\delta C(x, t)$: local change in coherence state
- $D(x_1, x_2)$: metric distance in emergent spacetime
- c_{eff} : effective projection velocity ($c_{\text{eff}} \leq c$)

We define the photon event Φ as a state update operator:

$$\Phi(x_1, x_2, t) = \delta C(x_1, t) \leftrightarrow \delta C(x_2, t)$$

where the bidirectional mapping $\leftrightarrow$ is instantaneous in the Bulk but bounded in emergent spacetime by c_{eff} .

The propagation time between x_1 and x_2 is thus:

$$\tau = D(x_1, x_2) / c_{\text{eff}}$$

6. Conclusion

By reframing photons as state update operators rather than energy packets, the Oyster Theory accommodates both the Bulk-instantaneous and spacetime-bounded aspects of information transfer. This provides a coherent bridge between high-level coherence dynamics and the observed limitations of relativistic physics, potentially opening new pathways for interpreting light propagation anomalies in cosmology.

Physical Interpretation of c_{eff} as an Emergent Projection Velocity

Within the framework of the Oyster Theory (OT), the effective speed of light c_{eff} is not treated as a fixed universal constant, but as an emergent property that reflects the coherence state and purity of the Hyperbrane in a given region of the Universe-Bubble.

While in our local environment $c_{\text{eff}} \approx c$ (the measured vacuum speed of light), variations may arise in regions where the Hyperbrane's coherence is degraded or modulated by external factors such as interactions with high-density matter, gravitational wells, or topological defects at the Bulk–Hyperbrane interface.

From an experimental perspective, c_{eff} could be quantified by detecting anomalies in photon propagation times between well-defined spatial points, particularly in astrophysical environments under extreme conditions: near black holes, within galaxy cluster filaments, or across voids with exceptionally low baryonic density. Such measurements would require extremely precise time-of-flight experiments, possibly leveraging long-baseline interferometry or pulsar timing arrays, comparing observed delays with predictions from standard cosmology.

Conceptually, if c_{eff} varies across environments, this would imply that local causality and the structure of emergent time are subtly dependent on the Hyperbrane's physical state. A lower c_{eff} would mean that the same Bulk-synchronous photon event $\Phi(x_1, x_2, t)$ would project into the Universe-Bubble with a longer measured latency τ , effectively stretching the perceived causal interval between events. In extreme cases, such variations could accumulate over cosmological scales, introducing measurable distortions in cosmological distance ladders or in the synchronization of distant reference frames. This interpretation challenges the traditional assumption of an invariant speed of light in vacuum, suggesting instead that the "vacuum" itself is a state-dependent projection from the Bulk via the Hyperbrane. In this sense, c_{eff} becomes a probe of the Hyperbrane's physical condition — a macroscopic order parameter linking the microphysics of resonance to the macroscopic geometry of spacetime.

Critical Perspectives on the Mapping of the Primordial Bulk into π 's Decimal Expansion

1. Normality of π

The universal and complete mapping relies on the conjecture of the normality of π , i.e., that all possible finite subsequences of digits appear with uniform frequency in its decimal expansion. This property has not yet been rigorously proven, representing a strong mathematical assumption on which the theoretical construction depends. A relaxed approach can be adopted by considering only the relevant subsequences for physically coherent universes, but it remains a significant limitation.

2. Formalization of the Isomorphism

At present, there is no operational and explicit definition to translate relational configurations of the Bulk into sequences of π 's digits, e.g., through univocal encodings of matrices or informational structures into decimal strings. The absence of a detailed algorithmic scheme hinders verification and validation tests.

3. Temporal Dynamics from a Static Archive

The infinite sequence of π is static and atemporal, while time in the bubble-universe is an emergent phenomenon. There is the conceptual and formal challenge of defining temporal progression as a coherent "reading index" in π 's sequence—an aspect crucial but still to be rigorously developed.

4. Extension to Multiple Constants or Multi-Bubble Models

The current model considers π as the sole archive; however, open ontological questions remain as to whether other mathematical constants (such as ϕ , e , or $\zeta(3)$) should be considered as parallel archives, forming a broader "library" of universes. It is not yet clear why, ontologically, our reality would emerge from π and not from other constants.

5. Strong Ontological Role of π

Assigning π the role of both Super-Attractor and absolute archive entails profound and speculative implications that require further argumentation and formalization to surpass the threshold of mere metaphysics and become a falsifiable scientific model. In summary, the main critical points concern the reliance on as-yet unproven mathematical properties (normality of π), the need for more explicit and operational formalizations, and the conceptual challenges linked to the emergence of time and the possible plurality of informational archives.

These aspects represent the most important theoretical and operational frontiers to address in developing the Oyster Theory into a robust and testable scientific structure.